

Scalable Spirituality: A Generative AI Workflow for Embodied Experience of Relief Heritage

Junming Dai
Laboratory of Art and Archaeology
Image, Ministry of Education
Zhejiang University
Hangzhou, China
2271654918@qq.com

Xiaojiao Chen
School of Art and Archaeology
Zhejiang University
Hangzhou, China
chenxiaojiao@zju.edu.cn

Fanjin Meng
Laboratory of Art and Archaeology
Image, Ministry of Education
Zhejiang University
Hangzhou, China
mengfanjin2024@163.com

Yonghao Chen
Laboratory of Art and Archaeology
Image, Ministry of Education
Zhejiang University
Hangzhou, China
12221223@zju.edu.cn

Xin Ge
Zhejiang University
Hangzhou, China
xingge@zju.edu.cn

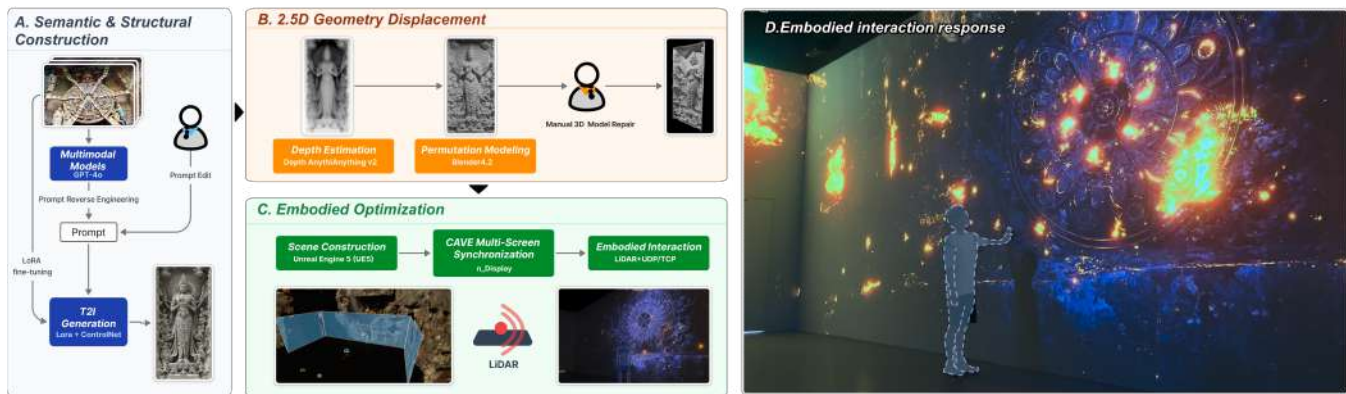


Figure 1: Overview of the proposed human-AI workflow for relief-based cultural heritage.

Abstract

Digital heritage experiences are shifting from static documentation toward participatory and embodied engagement, yet generative AI still struggles to produce geometrically faithful 3D assets for relief-based heritage, and many interactive tours reduce visitors to passive viewers. Using the Dazu Rock Carvings as a case, we present Scalable Spirituality, a workflow combining generative AI and LiDAR-based depth sensing to rapidly reconstruct and stage digital reliefs for immersive experiences. We employ a series of generative models to transform rock-carving images into 3D assets; to mitigate global warping and structural artifacts, we represent reliefs as 2.5D displacement geometry anchored by depth priors. We deploy the reconstructed reliefs in a CAVE, where real-time lighting is modulated by visitor proximity and tactile input to enhance embodied interaction. We show how human-AI collaboration can

scale relief reconstruction and enable experience-driven heritage presentation, reporting findings from a deployment and expert evaluation.

CCS Concepts

• **Human-centered computing** → **Mixed / augmented reality**; *Interactive systems and tools*; • **Applied computing** → *Arts and humanities*.

Keywords

Generative AI, Cultural Heritage, Embodied Interaction, Human-AI Collaboration, Depth-Based Reconstruction



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3798701>

ACM Reference Format:

Junming Dai, Xiaojiao Chen, Fanjin Meng, Yonghao Chen, and Xin Ge. 2026. Scalable Spirituality: A Generative AI Workflow for Embodied Experience of Relief Heritage. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3798701>

1 Introduction

Heritage digitization excels at preserving visual form but often misses the experiential and situated meaning that emerges through mediated engagement [2]. While Generative AI (GenAI) can enhance visual expressiveness, current implementations face a dual challenge. First, regarding fidelity, end-to-end 3D models frequently suffer from geometric hallucinations in complex reliefs, compromising historical authenticity [3]. Second, regarding interaction, despite the known value of bodily engagement in meaning-making [11], most systems confine users to passive viewing, failing to integrate their physical presence into the digital experience.

To address these gaps, we present Scalable Spirituality, a Human-AI collaborative workflow using the “Six Paths of Rebirth” reliefs at the Dazu Rock Carvings as a design implementation. We employ a depth-based 2.5D displacement strategy [14] to reconstruct stable geometry from single-view imagery, effectively mitigating structural distortions. Additionally, a LiDAR-based sensing layer maps visitor proximity and tactile input to real-time lighting and particle dynamics across the digital heritage asset and its ambient environment. This approach transforms the heritage site from a static object of gaze into a responsive, dynamic environment.

This paper contributes:

- **Scalable Spirituality Workflow:** A Human-AI collaborative pipeline integrating 2.5D depth-based reconstruction with LiDAR-driven embodied interaction. This workflow addresses the trade-off between geometric fidelity and real-time responsiveness, providing a replicable model for participatory relief heritage experiences.
- **Evaluation and Critical Reflection:** Insights derived from a field deployment, integrating qualitative design reflection, expert interviews, and visitor feedback. These findings validate the workflow’s feasibility and offer critical implications for balancing algorithmic expression with cultural authenticity.

2 Related Work

2.1 Generative AI in Heritage: Authenticity and Geometric Reconstruction

Generative AI is reshaping the construction of heritage narratives. He et al. [4] demonstrate that co-creating narratives with AI strengthens public emotional connections to history. However, its application in professional practice is constrained by reliability challenges. Witte et al. [12] note that current models often lack the cultural sensitivity required for conservation guidance, while Rathkopf [9] warns that pervasive “hallucinations” risk corroding scientific accuracy.

This tension between expressive potential and factual reliability is particularly acute in 3D reconstruction. While end-to-end text-to-3D models (e.g., TripoSR [10]) accelerate asset production, Condorelli [3] identifies that they frequently suffer from spatial hallucinations and geometric distortions, compromising iconographic authenticity. In response, Pannone et al. [7] suggest that for relief art, preserving high-frequency surface details is perceptually more critical than volumetric completeness. Consequently, building on

foundational work in robust monocular depth estimation [8], advanced models like Depth Anything V2 by Yang et al. [14] offer a viable 2.5D alternative. This approach balances generative efficiency with the structural fidelity required for authentic heritage documentation.

2.2 Embodied Cognition and Digital Heritage Experience

Theories of embodied cognition emphasize the role of bodily engagement in meaning-making. Trajkova et al. [11] establish that whole-body interaction significantly improves museum engagement compared to passive visual exploration, aligning with Nacke and Lindley’s [6] metrics on the relationship between interaction mechanics and immersion. Extending this to digital mediums, Wu et al. [13] validate the efficacy of VR gestures in reconstructing intangible rituals. Furthermore, Chen et al. [2] frame digital heritage meaning not as static display, but as an emergent property of “mediated engagement.”

3 Workflow Overview

To address the dual challenges of algorithmic hallucination and experiential disconnection in heritage digitization, we propose the Scalable Spirituality workflow. As illustrated in Fig. 1, the system architecture comprises three tightly coupled phases: (1) Semantic Construction, establishing iconographic boundaries; (2) 2.5D Depth-Based Displacement, ensuring geometric topological stability; and (3) LiDAR-Driven Embodied Integration, facilitating the shift from static display to dynamic interaction.

3.1 Semantic Construction

As a prerequisite for reconstruction, we employ a semantically constrained synthesis strategy (Fig. 1A). Archaeological imagery is first encoded into structured prompts rich in cultural semantics to guide text-to-image generation. The synthesis utilizes diffusion models constrained by ControlNet [15] to maintain compositional consistency. Concurrently, LoRA-based iconographic constraints [5] are applied to regulate symbolic elements (e.g., *mudrā* and ritual implements), effectively suppressing uncontrolled Iconographic Distortions

3.2 2.5D Depth-Based Displacement

To mitigate the geometric distortions typical of end-to-end 3D models [3], we adopt a 2.5D depth-based displacement strategy (Fig. 1B). Integrated within a customized ComfyUI workflow, the pipeline utilizes Depth Anything V2 [14] to infer high-resolution depth maps from single-view imagery. The model’s robustness to complex textures allows for the precise capture of high-frequency surface details. Subsequently, a Human-in-the-Loop refinement step corrects depth artifacts caused by lighting ambiguities, mapping depth to geometric displacement to preserve perceptual realism while preventing topological collapse.

3.3 LiDAR-Driven Embodied Integration

Aligning with Trajkova et al.’s [11] emphasis on bodily engagement, the optimized assets are deployed in an Unreal Engine 5-based CAVE immersive environment (Fig. 1C). The system employs nDisplay for synchronized multi-screen rendering and Lumen for real-time global illumination, simulating the low-light atmosphere characteristic of grottoes. To accommodate these conditions, an RPLIDAR A2 sensor serves as the primary sensing layer. Distinct from symbolic gesture recognition, this framework transmits visitor proximity and tactile input via UDP, mapping them directly to real-time lighting and particle dynamics. This design transforms bodily presence into a driver of environmental evolution, shifting the experience from static viewing to an embodied dialogue.

4 Design Implementation: “Digital Samsara” of the Dazu Rock Carvings

To validate the proposed workflow, this paper presents a full-scale design case, “Digital Samsara,” based on the UNESCO-listed Dazu Rock Carvings. Focusing on the narrative “Six Paths of Rebirth” reliefs, the installation reinterprets the Buddhist cycle of existence as a fluid digital regeneration. This case translates the generative pipeline into a responsive spatial experience, providing a concrete basis for field evaluation.

4.1 Spatial Instantiation of Cyclical Temporality

The experience space adopts a circular topology rather than a linear exhibition layout, reflecting the cyclical temporality of Samsara. This spatial organization allows visitors to experience repetition and return through bodily movement, instead of following a fixed narrative sequence. At the asset level, relief geometry is reconstructed using the depth-based 2.5D displacement pipeline described earlier. By constraining geometry generation to a single calibrated viewpoint, large-scale structural hallucinations common in end-to-end 3D generation are avoided. Ambiguous depth regions—such as shadow-induced concavities—are further corrected through manual adjustment in Blender, ensuring topological plausibility via Human-in-the-Loop refinement.

Based on the temporal characteristics of Samsara, four sets of visual parameters are defined within the real-time rendering workflow to cyclically modulate the environment over time (Figure 2):

- **Birth (Dawn, Fig. 2a):** High-intensity diffuse lighting combined with low-saturation cyan tones produces a low-contrast environment in which surface details remain continuous but understated.
- **Prime (Noon, Fig. 2b):** Stronger directional lighting, increased contrast, and warmer color temperatures amplify surface normal response and specularities, emphasizing relief depth and material granularity.
- **Aging (Dusk, Fig. 2c):** Overall illumination intensity is reduced and shifted to lower angles, allowing shadow regions to expand and merge, perceptually destabilizing previously distinct forms.
- **Rebirth (Void, Fig. 2d):** Most direct illumination is removed, leaving only low-frequency, pulsating emissive traces embedded in the geometry as transitional cues between cycles.

Table 1: Comparison of reconstruction strategies based on standard digital heritage metrics.

Metric	End-to-End Gen.	3D 2.5D Displacement (Ours)
Input Priors	Unposed	Calibrated
Topological Regularity	Non-manifold	Regular Grid
Geometric Fidelity	Hallucinated	Source-Aligned
Generative Stability	Stochastic	Deterministic
Asset Usability	Baked / Rigid	Decoupled
Primary Artifacts	Janus Problem	View Distortion
Target Typology	Freestanding	Reliefs

All visual transitions are driven by real-time rendering rather than pre-authored animation, ensuring sustained system operation in the deployed environment.

4.2 Continuous Embodied Interaction without Symbolic Encoding

Embodied interaction is implemented as continuous modulation rather than symbolic control. Using LiDAR sensing, visitors’ proximity and movement are mapped directly to gradual changes in lighting intensity and particle behavior, integrating bodily presence the environment without predefined interaction scripts.

Sustained proximity produces slow accumulative effects, while abrupt movements introduce temporary disturbances that dissolve and reassemble visual structures. This contrast allows embodied engagement to shape the environment while maintaining system stability, as shown in Fig. 1D.

5 Evaluation

5.1 Comparative Analysis of Reconstruction Strategies for Relief Heritage

This paper conducted a qualitative comparative analysis between end-to-end 3D generation approaches and the proposed 2.5D depth-based displacement strategy in the context of relief-based cultural heritage. While end-to-end models offer advantages in generation speed and viewpoint flexibility, their outputs often exhibit unstable mesh topology and geometric artifacts, particularly in artifacts with complex structures such as multi-armed figures, where redundant limbs or over-smoothed erosion details are frequently observed.

In contrast, the proposed 2.5D strategy prioritizes geometric fidelity and generative stability by anchoring reconstruction to the input’s perspective priors. While inherently view-dependent, this approach ensures the preservation of high-frequency relief details. Furthermore, its topological regularity significantly enhances asset usability, facilitating precise post-processing and control. Table 1 synthesizes the qualitative comparison between these strategies based on critical metrics for heritage digitization.

5.2 Visitor Feedback

This paper collected exploratory feedback from visitors who experienced the installation during public exhibition sessions. Participation was voluntary and anonymous, and feedback was gathered

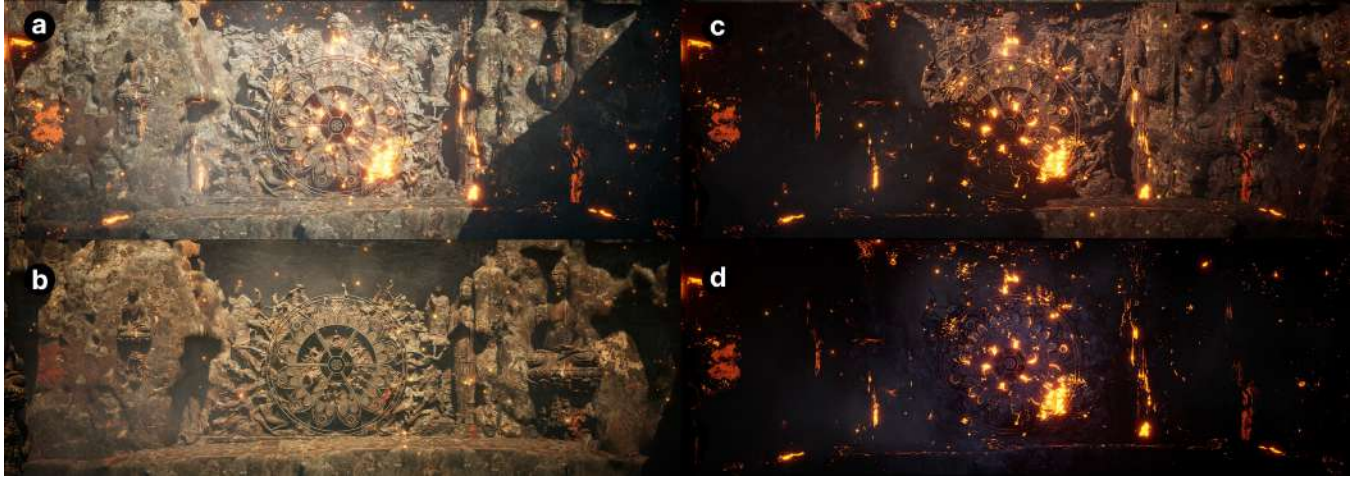


Figure 2: Time-cyclic rendering states of the immersive relief environment.

Table 2: Summary of visitor feedback (7-point Likert scale; 1 = strongly disagree, 7 = strongly agree).

Statement	$M \pm SD$
I could clearly perceive how the environment responded to my bodily position or movement	6.24 ± 0.94
The system's responses enhanced the overall sense of immersion	6.35 ± 1.05
The experience felt conceptually coherent with the theme of cyclic temporality	6.29 ± 0.93
The interaction felt natural and comfortable in a low-light exhibition environment	6.29 ± 0.90

using a short post-hoc questionnaire combining 7-point Likert-scale items (1 = strongly disagree, 7 = strongly agree) and open-ended prompts. Responses were analyzed descriptively to summarize recurring experiential observations.

A total of $N=17$ visitors completed the questionnaire.

Across post-hoc questionnaire responses (Table 2), participants reported high agreement that the environment responded clearly to their bodily position and movement ($M = 6.24, SD = 0.94$), contributing to an enhanced sense of immersion ($M = 6.35, SD = 1.05$). Ratings also indicated that the interaction remained comfortable in low-light exhibition conditions ($M = 6.29, SD = 0.90$) and conceptually coherent with the theme of cyclic temporality ($M = 6.29, SD = 0.93$). Informal post-experience comments further suggested that the visual presentation was perceived as visually striking, reinforcing the overall experiential impact.

5.3 Expert Interviews

To complement visitor feedback, this paper conducted brief semi-structured interviews with three domain experts. The interviews were summarized descriptively as qualitative triangulation rather than a formal evaluation.

The heritage/museum expert stressed that semantically constrained synthesis is acceptable only when its boundaries are explicit and auditable, and that Human-in-the-Loop review should document what is constrained versus stylistically inferred. The computer science expert noted that 2.5D displacement yields a regular planar grid that benefits real-time deployment, but recommended clearly reporting view-dependence and typical failure cases (e.g., shadow-induced depth ambiguity). The art and design expert highlighted that the temporal cycle and breathing-like illumination create a legible embodied rhythm, while suggesting tighter alignment between cycle phases and perceived affect to strengthen narrative-free coherence in low-light settings.

6 Discussion and Conclusion

This paper introduces the Scalable Spirituality workflow for creating interactive digital experiences for relief-based cultural heritage. By suppressing geometric distortion and coupling bodily movement with visual generation, the workflow reactivates experiential and spiritual dimensions of heritage, shifting it from static preservation to dynamic perception. Rather than pursuing fully automated or free-viewpoint 3D reconstruction, it focuses on frontal-view relief heritage, balancing technical feasibility with cultural authenticity. Within this scope, the approach improves structural stability, surface fidelity, and process controllability, while clearly defining its limitations for Complex volumetric artifacts. Although the human-in-the-loop process relies on expert judgment, such involvement is necessary in context-sensitive heritage interpretation [1].

Several limitations of this work warrant discussion. Current LiDAR interaction responds only to coarse spatial movement and does not integrate fine-grained multimodal inputs. In addition, the case study addresses a single type of relief heritage, and its applicability to other forms remains unverified. Future work may explore lightweight generative models with hardware acceleration, multimodal language models for dialogic interpretation, and systematic evaluation in museum and exhibition contexts.

In summary, this work integrates generative AI into embodied experience design. Through the Digital Samsara of Dazu Rock

Carvings case, we show that generative embodied experiences can transform cultural heritage from static archives into perceivable and participatory cultural processes, offering new ways to sustain spiritual meaning in the digital age.

Acknowledgments

The authors would like to thank Professor Xiaojiao Chen for her guidance and mentorship throughout this research. This work was supported by the Major Project of the Zhejiang Provincial Philosophy and Social Sciences Research Base (No. 2025JDKT18Z) and the China National Arts Fund (No. 2025-A-04-092-490).

References

- [1] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N. Bennett, Kori Inkpen, Jaime Teevan, Ruth Kikin-Gil, and Eric Horvitz. 2019. Guidelines for Human-AI Interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3290605.3300233
- [2] Rongyi Chen, Ziyang Xin, Qing Xiao, Ruiwei Xiao, Jingjia Xiao, Bingbing Zhang, Hong Shen, and Zhicong Lu. 2025. The Digital Landscape of God: Narrative, Visuals and Viewer Engagement of Religious Videos on YouTube. arXiv:2509.10957 [cs.HC] <https://arxiv.org/abs/2509.10957>
- [3] Francesca Condorelli. 2025. *3D Models from Text Descriptions: Using Artificial Intelligence for Representation of Cultural Heritage*. FrancoAngeli srl, Milan, Italy. doi:10.3280/oa-1430-c893
- [4] Zhiting He, Jiayi Su, Li Chen, Tianqi Wang, and Ray Lc. 2025. 'I Recall the Past': Exploring How People Collaborate with Generative AI to Create Cultural Heritage Narratives. *Proc. ACM Hum.-Comput. Interact.* 9, 2, Article CSCW108 (May 2025), 30 pages. doi:10.1145/3711006
- [5] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. LoRA: Low-Rank Adaptation of Large Language Models. arXiv:2106.09685 [cs.CL] <https://arxiv.org/abs/2106.09685>
- [6] Lennart Nacke and Craig A. Lindley. 2008. Flow and immersion in first-person shooters: measuring the player's gameplay experience. In *Proceedings of the 2008 Conference on Future Play: Research, Play, Share* (Toronto, Ontario, Canada) (Future Play '08). Association for Computing Machinery, New York, NY, USA, 81–88. doi:10.1145/1496984.1496998
- [7] Daniele Pannone, Alessia Castronovo, Maurizio Mancini, Gian Luca Foresti, Claudio Piciarelli, Rossana Gabrieli, Muhammad Yasir Bilal, and Danilo Avola. 2026. Augmented Reality in Cultural Heritage: A Dual-Model Pipeline for 3D Artwork Reconstruction. In *Image Analysis and Processing – ICIAP 2025: 23rd International Conference, Rome, Italy, September 15–19, 2025, Proceedings, Part II* (Rome, Italy). Springer-Verlag, Berlin, Heidelberg, 274–285. doi:10.1007/978-3-032-10192-1_23
- [8] René Ranftl, Katrin Lasinger, David Hafner, Konrad Schindler, and Vladlen Koltun. 2020. Towards robust monocular depth estimation: Mixing datasets for zero-shot cross-dataset transfer. *IEEE transactions on pattern analysis and machine intelligence* 44, 3 (2020), 1623–1637.
- [9] Charles Rathkopf. 2026. Hallucination, reliability, and the role of generative AI in science. arXiv:2504.08526 [cs.CY] <https://arxiv.org/abs/2504.08526>
- [10] Dmitry Tochilkin, David Pankratz, Zexiang Liu, Zixuan Huang, Adam Letts, Yangguang Li, Ding Liang, Christian Laforte, Varun Jampani, and Yan-Pei Cao. 2024. TripoSR: Fast 3D Object Reconstruction from a Single Image. arXiv:2403.02151 [cs.CV] <https://arxiv.org/abs/2403.02151>
- [11] Milka Trajkova, A'eshah Alhakamy, Francesco Cafaro, Rashmi Mallappa, and Sreekanth R. Kankara. 2020. Move Your Body: Engaging Museum Visitors with Human-Data Interaction. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3313831.3376186
- [12] Jessica Witte, Edmund Lee, Lisa Brausem, Verity Shillabeer, and Chiara Bonacchi. 2025. Generative AI in Heritage Practice: Improving the Accessibility of Heritage Guidance. *Heritage* 8, 12 (2025). doi:10.3390/heritage8120513
- [13] Yiping Wu, Yue Li, Eugene Ch'ng, Jiaxin Gao, and Tao Hong. 2025. KaiBiLi: gesture-based immersive virtual reality ceremony for traditional Chinese cultural activities. *Visual Computing for Industry, Biomedicine, and Art* 8, 1 (02 Oct 2025), 24. doi:10.1186/s42492-025-00205-x
- [14] Lihe Yang, Bingyi Kang, Zilong Huang, Zhen Zhao, Xiaogang Xu, Jiashi Feng, and Hengshuang Zhao. 2024. Depth anything V2. In *Proceedings of the 38th International Conference on Neural Information Processing Systems* (Vancouver, BC, Canada) (NIPS '24). Curran Associates Inc., Red Hook, NY, USA, Article 688, 37 pages.
- [15] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. 2023. Adding Conditional Control to Text-to-Image Diffusion Models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 3836–3847.